

My Vivino — Data-Driven Wine Recommendation System



Replacing a rules-based engine with scalable machine learning

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Business Problem



The Challenge:

- My Vivino serves thousands of wines to a large user base
- Existing recommendations rely on static rules
- Limited personalization reduces engagement and discovery

Objective

Develop a scalable recommendation system that leverages historical user behavior to deliver personalized wine suggestions.

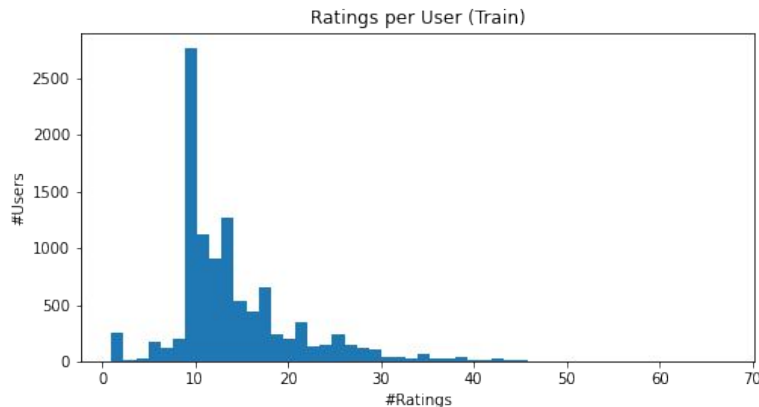
Strategic Shift

Rules → Data → Personalization

Dataset & Modeling Rationale

Dataset Overview — X-Wines:

- 1,007 wines
- 10,561 users
- 150,000 historical ratings



Key Insight

The highly sparse user-item interaction matrix motivates the use of matrix factorization-based collaborative filtering.

Recommendation Examples

RMSE (CF): 0.6800

CF recommendations for UserID=1209683

	WineID	WineName	Type	Country	avg_rating	num_ratings	harmonize_norm
0	111495	Pauillac (Grand Cru Classé)	Red	France	4.219190	568	[beef, lamb, game meat, poultry]
1	155443	Cava Clasico Brut	Sparkling	Spain	3.335737	624	[shellfish, appetizer, snack, lean fish, cured...]
2	112084	Oenothèque Brut Champagne	Sparkling	France	4.757212	208	[pork, rich fish, shellfish, soft cheese]
3	167418	Malbec	Red	Argentina	3.879870	1540	[beef, lamb, poultry]
4	180330	IX Estate Red	Red	United States	4.728261	92	[beef, lamb, game meat, poultry]

CF + Food='beef' recommendations for UserID=1209683

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1	167418	Malbec	Red	Argentina	3.879870	1540	[beef, lamb, poultry]
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3	174333	The Factor	Red	Australia	4.369231	195	[beef, lamb, game meat, poultry]
4	111516	Haut-Médoc (Grand Cru Classé)	Red	France	4.199597	496	[beef, lamb, game meat, poultry]

What This Shows:

- Personalized discovery
- Context-aware suggestions
- Production-ready fallback

Business Impact



Expected Impact

- Improved recommendation
- Increased user engagement
- Higher conversion potential
- Better product discovery

Production Readiness

- Modular pipeline
- No data leakage
- Robust cold-start strategy
- Scalable architecture

Strategic Value

Establishes the foundation for real-time personalization at scale.